

A Better Automated Market Maker

See how real AMMs actually behave — then *design and try* a better one, live.

 amm.finance

Ernesto Sanches · everything in this deck is real and reproducible from the repo

Liquidity you can't see, can't price

- **LPs** earn less than the risk they take — capital sits where it never trades.
- **Traders** pay more than they should — wider spreads, more slippage.
- An AMM curve is an **invisible, abstract object**. Nobody has good intuition for what a liquidity shape *does*.

You can't improve what you can't see, measure, or safely experiment with.

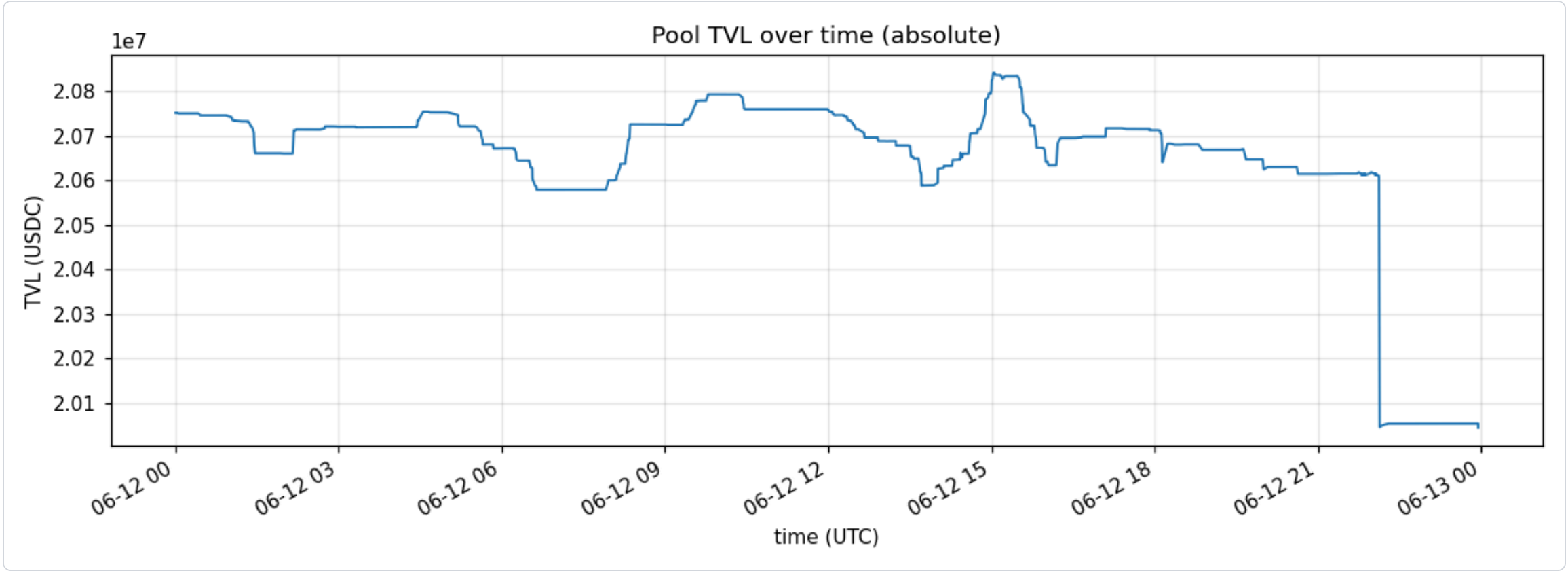
An AMM is an order book

Bids below spot (green), asks above (red). A swap slides the line — and every level it crosses flips side.



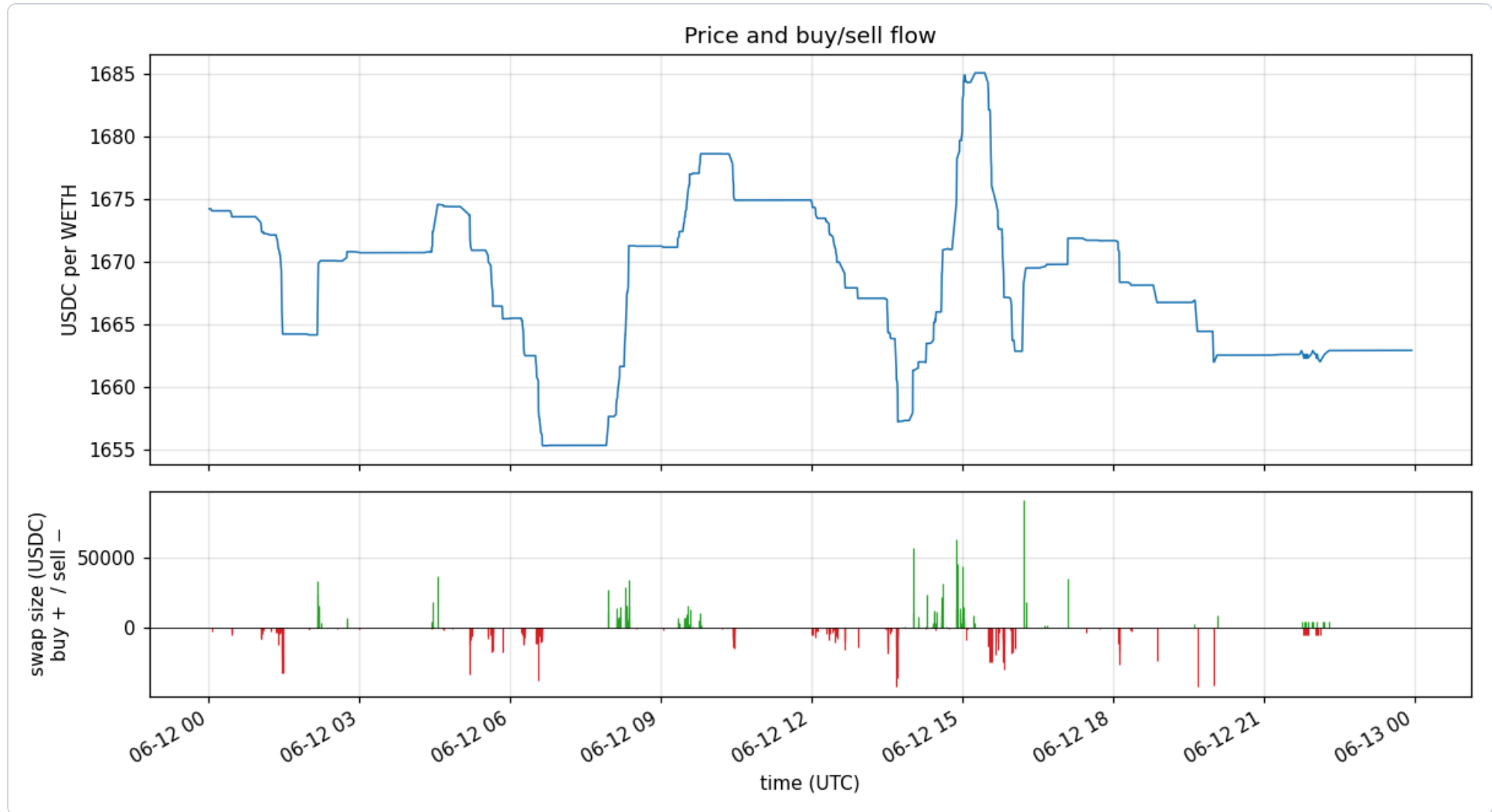
One real pool, one real day — reconstructed with no account

ETH/USDC 0.3%, 24h of TVL — keyless public RPC only: no wallet, no API key, no Graph/Dune.



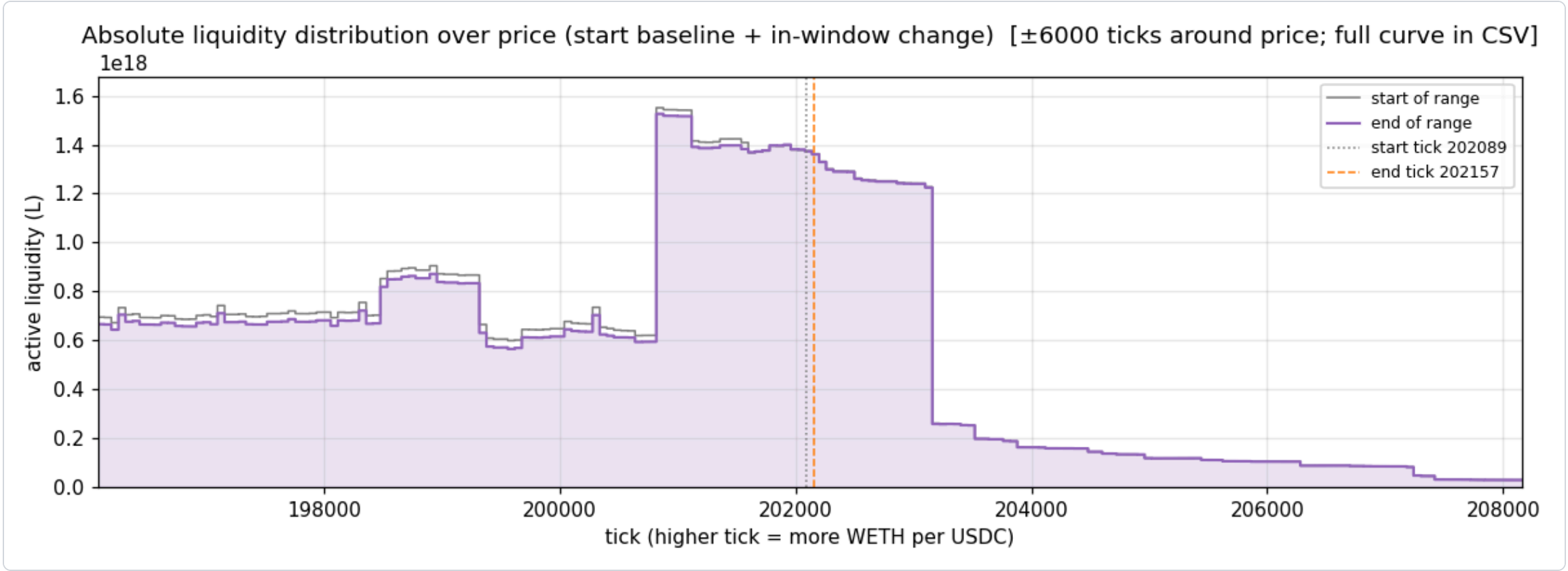
Every trade, classified — buys up, sells down

770 swaps across the day, each priced and sided straight from the chain.



Where the liquidity actually sits

Absolute depth across price, start vs end of day — recovered from on-chain state, not estimated.



...decomposed into individual LP orders

Each color is one liquidity provider's position, replayed across the whole day with a time slider.



It checks out against reality

\$2.73M volume · \$8.2k fees · **14.3% APR**

Computed by replaying every swap through our engine — and it lands squarely on Uniswap's published number for this pool. The model is validated against the real thing.

Trade now on our AMM!

Compete with your peers here right now → amm.finance

AMM Game

LOBBY

D = 3,000 USD0/ETH0

60:00 left

Play

Leaderboard

Profile

Admin

as **alice**

Portfolio

USD0

\$5,000

ETH0

1.6667 (\$5,000)

In LP positions

\$0

Total value

\$10,000

External price (D, USD0/ETH0)

3,000

Untradeable mark. Pushing a pool toward D is a directional bet.

v3 pool — range liquidity

detail →

Pool price

3,000 USD0/ETH0

TVL

\$0

Your fees

\$0

Trading is lobby.

USD0 in

Buy ETH0

ETH0 in

Sell ETH0

► Deposit liquidity

No positions in this pool.

curve pool — draw your curve

detail →

Pool price

3,000 USD0/ETH0

TVL

\$0

Your fees

\$0

Trading is lobby.

USD0 in

Buy ETH0

ETH0 in

Sell ETH0

► Deposit liquidity

No positions in this pool.

connected · server-authoritative · vanilla JS, no build step

Uniswap's AMM vs. our new design — side by side

Left pool is classic Uniswap v3. Right pool is our model. Trade and provide liquidity on both, and watch which one serves traders and LPs better.

AMM Game

RUNNING

D = 3,000 USD0/ETH0

60:00 left

Play

Leaderboard

Profile

Admin

as **alice**

Portfolio

USD0

\$4,500

ETH0

1.8291 (\$5,487)

In LP positions

\$0

Total value

\$9,987

v3 pool — range liquidity

detail →

Pool price

3,138.03 USD0/ETH0

TVL

\$30,679

Your fees

\$0

500

Buy ETH0

ETH0 in

Sell ETH0

▶ Deposit liquidity

No positions in this pool.

External price (D, USD0/ETH0)

3,000

Untradeable mark. Pushing a pool toward D is a directional bet.

curve pool — draw your curve

detail →

Pool price

3,000 USD0/ETH0

TVL

\$30,000

Your fees

\$0

USD0 in

Buy ETH0

ETH0 in

Sell ETH0

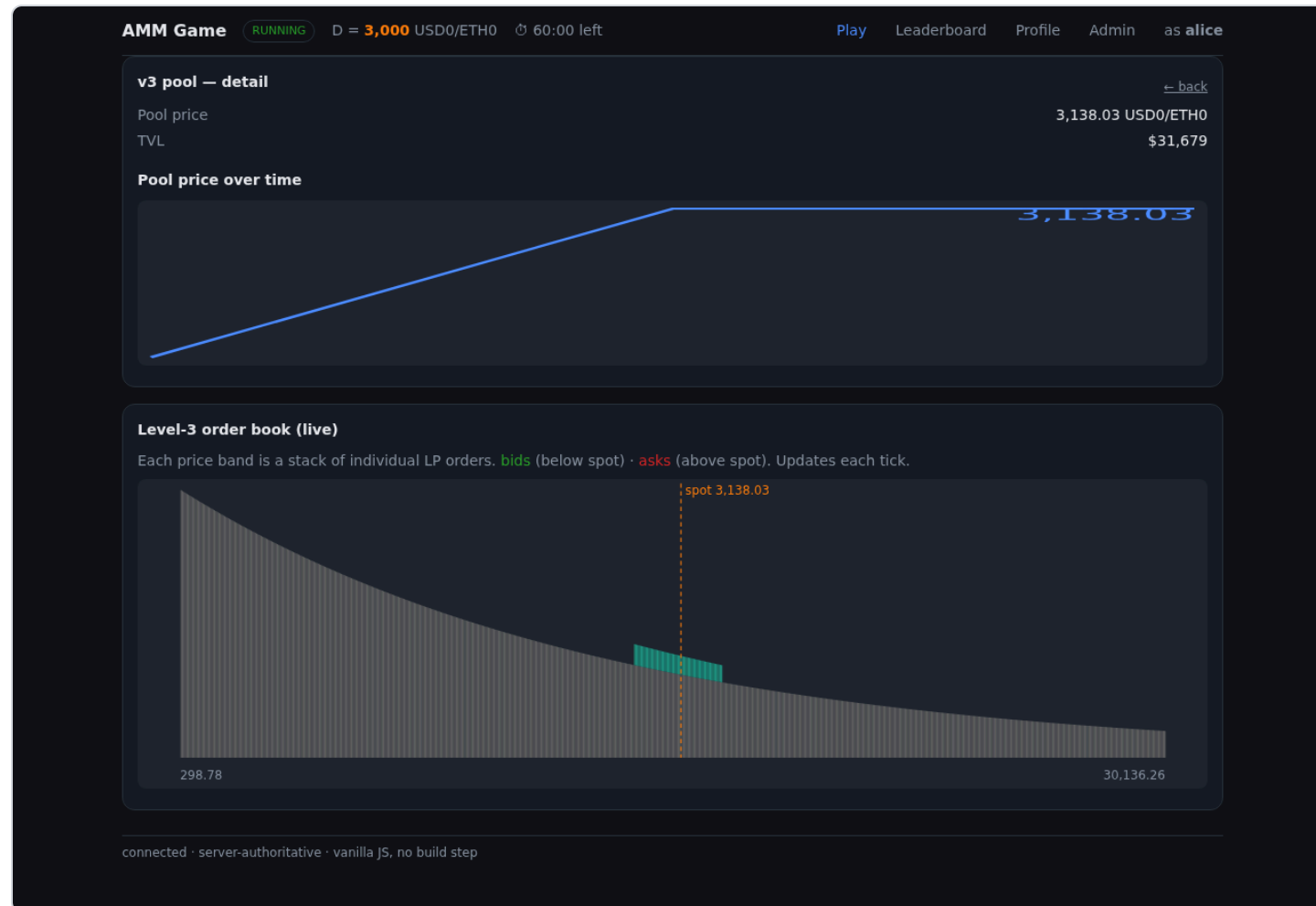
▶ Deposit liquidity

No positions in this pool.

connected · server-authoritative · vanilla JS, no build step

Our model: you draw the AMM curve yourself

Instead of Uniswap's fixed price range, place liquidity in *any* shape you want — it becomes a live order book (the colored hump on the grey baseline). A real way to experiment with better curves.



Which design earns more?

Every participant marked against a passive-LP benchmark — a direct, live experiment in whether a hand-drawn curve can beat Uniswap's fixed-range model.

AMM Game

RUNNING

D = 3,000 USD0/ETH0

⌚ 60:00 left

Play

Leaderboard

Profile

Admin

as **alice**

Leaderboard

Marked at current D = 3,000 USD0/ETH0. House rows are non-winning benchmarks.

#	Player	Total value	USD0	ETH0	Fees	Taker vol	Maker vol
1	alice	\$9,987	\$3,952	1.685	\$0	\$500	\$0
·	House (seed, v3)	\$10,004	\$0	0	\$0	\$0	\$0
·	House (seed, curve)	\$10,000	\$0	0	\$0	\$0	\$0

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Every experiment, fully tracked

Balances, fees, volume, open positions, full action history — and value-over-time, so each participant can see exactly how their AMM design performed.

AMM Game

RUNNING

D = 3,000 USD0/ETH0

🕒 60:00 left

Play

Leaderboard

Profile

Admin

as **alice**

Profile — **alice**

alice

Rename

USD0

\$3,846.12

ETH0

1.7188

Fees collected

\$0

Taker volume

\$500

Maker volume

\$0

Portfolio value over time (marked at D)

waiting for data...

Open positions

#3 v3/range

\$985

Action history

#4

deposit

{"kind":"range","tick_lower":77820,"tick_upper":81900,"budget_usd0":1000,"pool":

#3

buy

{"amount_in":500,"pool":"v3"}

#1

register

{"name":"alice","is_admin":false}

connected · server-authoritative · vanilla JS, no build step

Runs as a real event

One operator configures and starts it. The ledger reads exact to float-epsilon, and a mid-game crash restarts with byte-identical state — zero loss.

AMM Game

RUNNING

D = 3,000 USD0/ETH0

🕒 60:00 left

Play

Leaderboard

Profile

Admin

as **alice**

Parameters

Locked — game is running.

Initial price D₀

3000

σ per step

0.0045

walk_step (s)

5

pool fee (γ)

0.003

seed depth k

3

player bag X

10000

game length (s)

3600

size cap (frac)

0.1

range factor

10

base name

ETH0

quote name

USD0

Save params

Start game

Monitor

Phase

RUNNING

D

3,000

Time left

60:00

Conservation

OK (Δ 3.6e-16 / -2.5e-12)

2 accounts

Account	USD0	ETH0	Pos
alice	\$3,846	1.7188	1
House	\$2	0	2

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Try it:  amm.finance

Measure → `python run_all.py`

Try the new AMM → `python app/run.py`

We made the invisible AMM curve visible and measurable — then built a better one you can design and try for yourself.